

RAYMOND SHAW

346-526-9031 | raymondsh0705@gmail.com | www.linkedin.com/in/raymond-shaw-0baa67282
https://github.com/RaymondSh0705

EDUCATION

The University of Texas at Austin, Austin, TX *Expected Graduation: May 2028*
Bachelor of Science, Computer Science | Minors: Economics, Japanese | Current GPA: 3.96
Relevant Coursework: Speech/Audio Processing, Operating Systems, Computer Architecture, Data Structures

SKILLS

Technical Skills: Python, Java, C, SQL, JavaScript, PyTorch, Scikit-learn, Hugging Face, React, Node.js, AI/ML, LLM, Computer Vision, Data Analysis, Data Schema Design, Data Pipelining, Web Development, Chinese, Japanese

EXPERIENCE

- AMRL Lab**, Austin, TX | *Undergraduate Researcher* *September 2026 - Current*
- Researching Omniword model integration to assist in social navigation for autonomous robotic pathing.
 - Training a Cosmos-3 model to output socially acceptable primitives for ComposableNav diffusion planning.
 - Developing a custom SFT and heuristic evaluation pipeline and integrating with ROS-2 for robotic simulation.
- MetOX International**, Houston, TX | *AI/ML and Data Analyst Intern* *June 2026 - August 2026*
- Automated manufacturing processes with ML and developed data reasoning and modeling web applications.
 - Trained Yolo and PatchCore CV models to detect live manufacturing defects, with live-viewing app integration.
 - Produced schemas and ETL pipelines to transfer high-volume data between PLC signals and MySQL databases.
- McComb's AI/ML Business Application Program**, Online | *Program Participant* *December 2025 - September 2026*
- Studied deep learning, NLP, Computer Vision, and Gen. AI foundations through live technical mentoring.
 - Developed AI agents using frameworks such as PyTorch, NLTK, SKlearn, etc., to solve real-world business cases.
 - Implemented complete ML pipelines, including preprocessing, model training with SFT and RAG, and evaluation.
- MAG Neurospine Clinic**, Houston, TX | *Web Developer Intern* *July 2025 - August 2025*
- Developed a web app database to manage and securely access patient data entries into Microsoft Dataverse.
 - Automated the migration of legacy PDF and Excel data to Power Apps using SQL and Power Query.
 - Prepared technical reports to streamline communication with stakeholders on planning and execution.
- NASA High School Aerospace Scholars Program**, Houston, TX | *Project Manager* *September 2023 - June 2024*
- Partnered with NASA engineers and led a 10-person research group to simulate a lunar landing mission.
 - Integrated CAD modeling and programming simulations for lunar base prototype designs.

PROJECTS

- Kanadle - Web Application** *September 2026 - Current*
- Developing a multi-layered, responsive application using JavaScript and React for interactive UI and state logic.
 - Employs a Node.js backend integrating Jisho API requests to receive and structure dynamic game logic.
- Japanese Word Bomb - Online Web-based Multiplayer Game** *November 2025 - January 2026*
- Built a real-time Japanese word query multiplayer game using Python WebSockets, FastAPI, and Pydantic.
 - Designed custom network protocols, private lobby system, and player device sync for smooth gameplay.

ORGANIZATIONS

- DiRP**, Austin, TX | *LLM-Training Mentee* *January 2026 - Current*
- Applied transformer, RoPE, and tokenization techniques with PyTorch to develop and train a Nano-LLM Chatbot.
 - Studied and reproduced modern model innovations such as MoE architecture and memory pipelining.
- ECLAIR**, Austin, TX | *Development Member* *October 2025 - December 2025*
- Contributed to *Trace*, an automated rotating platform that moves a placed object's motion to follow a digital drawing's outline with motion tracking and live computer algorithm calculations.
 - Built a custom Python drawing interface and mapped drawings and velocities through Bezier curve modeling.
- Elkins HS Math and Science Club**, Missouri City, TX | *Vice President* *September 2023 - May 2025*
- Co-led and won with 4 state-level UIL/TMSCA competition math teams (first wins since 2004!).
 - Planned and taught bi-weekly training sessions and built custom TI-Nspire calculator functions.

AWARDS

- 1st place team - UIL State Competition (Mathematics and Calculator) *May 2025*
- 1st place team - TMSCA State Competition (Mathematics and Calculator) *March 2025*
- 1590 - SAT Score *December 2024*
- 2nd place, 3rd place - University of Houston Math Contest *February 2023, February 2024*